

1.1.8. Matrix Multiplication

13:35 A ↻ ↗ ⌂ -

Write a program to multiply two matrices A and B. The program should read their dimensions and elements, check whether matrix multiplication is possible, and display the resulting product matrix.

If the number of columns in matrix A is not equal to the number of rows in matrix B, the program should skip reading the elements of the second matrix and print "Invalid input".

Input Format:

- The first line contains two integers, m and n , representing the dimensions of matrix A.
- The next m lines each contain n space-separated integers, representing matrix A.
- The next line contains two integers, r and p , representing the dimensions of matrix B.
- The next r lines each contain p space-separated integers, representing matrix B.

Output Format:

- Print the resulting matrix after multiplication, where each row is displayed on a new line with space-separated integers.
- If multiplication is not possible (number of columns of A $\neq$ number of rows of B), print "Invalid input".

Sample Test Cases

+

Explorer

C CTC77533.c

Submit

Debugger

```
1 #include <stdio.h>
2 int main()
3 {
4     int r1, c1, r2, c2;
5     int A[100][100], B[100][100], C[100][100];
6     // Read Matrix A dimensions
7     scanf("%d %d", &r1, &c1);
8     // Read Matrix A
9
10
11
```

Average time
0.070 s
70.25 ms

Maximum time
0.076 s
76.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 76 ms Debug

Actual output

Expected output	Actual output
2 3	2 3
1 2 3	1 2 3
4 5 6	4 5 6
3 2	3 2
7 8	7 8
9 10	9 10

Terminal

Test cases

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